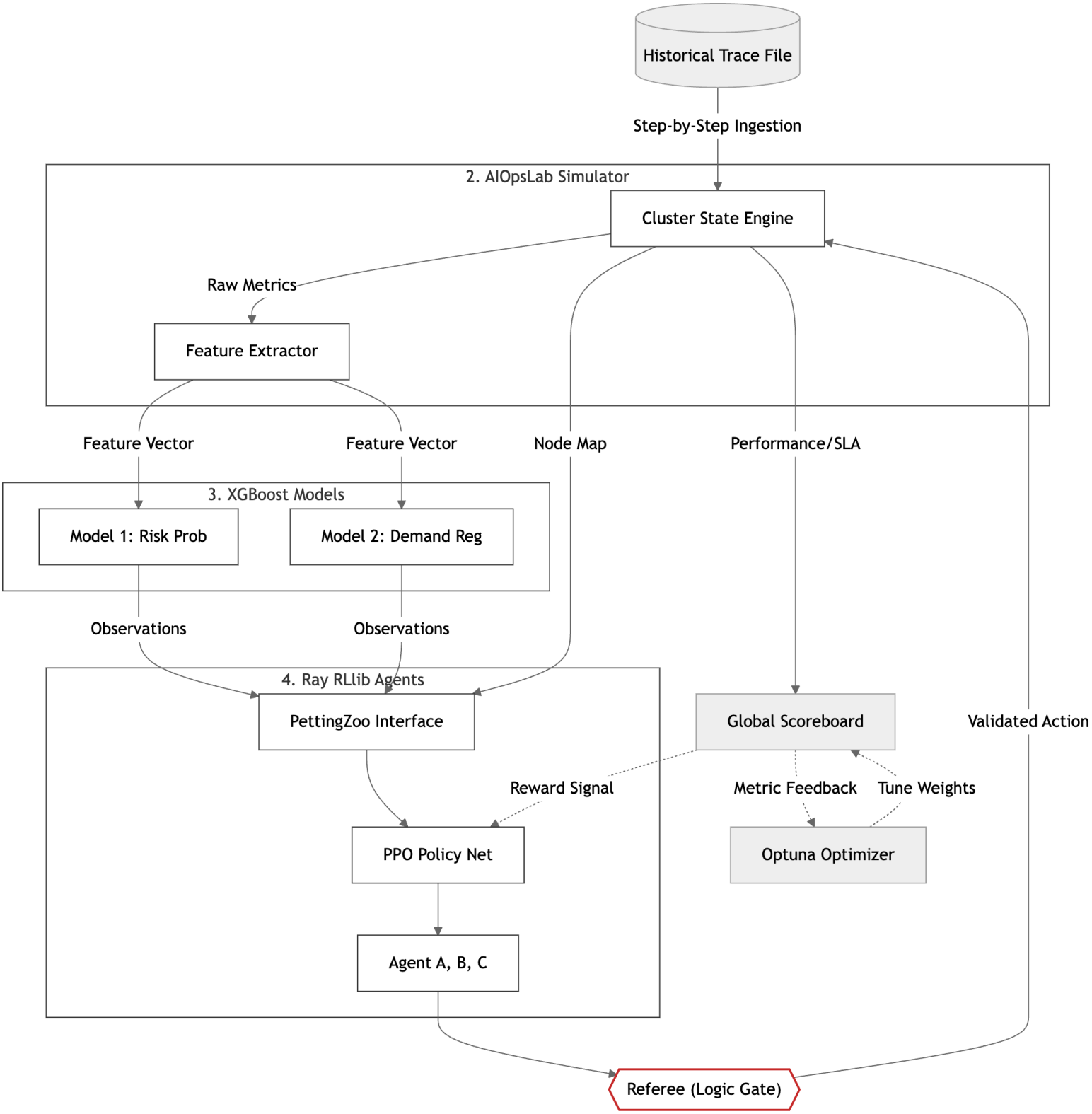
Historical Trace File
Step-by-Step Ingestion
2. AIOpsLab Simulator
Cluster State Engine
Raw Metrics
Feature Extractor
Feature Vector
Feature Vector
Node Map
Performance/SLA
3. XGBoost Models
Model 1: Risk Prob
Model 2: Demand Reg
Observations
Observations
4. Ray RLlib Agents
PettingZoo Interface
Global Scoreboard
Validated Action
Reward Signal
Metric Feedback
Tune Weights
PPO Policy Net
Optuna Optimizer
Agent A, B, C
Referee (Logic Gate)

Layer	Component	Tool / Language	Core Function	Interaction Flow	Logic / Data
1. Local Source	**Real Server Nodes**	Hardware / Linux	Generates real-world infra metrics	Prometheus collects data every X seconds	CPU, RAM, Disk, Network I/O
1. Local Source	**Prometheus / JSON**	Python	Converts metrics to digital history	Saves data as .csv or .json Traces	Time-series data for simulation
2. Simulator	**AIOpsLab (Twin)**	Docker / Python	The "Gym" for AI training	Ingests Traces; mimics node behavior	Executes actions & returns Rewards
2.5 Interface	**PettingZoo**	Python	**The Standard Bridge.** Wraps Sim into a Multi-Agent API	AIOpsLab → **PettingZoo** → Ray RLlib (PPO)	Observation/Action Space Mapping; Agent Turn-taking
3. Brains	**XGBoost 1 (Safety)**	Python / XGBoost	Predicts failure risk for tasks	Feeds risk scores to Agent A	Probability (Risk 0.0-1.0)
3. Brains	**XGBoost 2 (Resource)**	Python / XGBoost	Forecasts future resource demand	Feeds "Demand" scores to Agent B & C	Regression (Predicted resource usage)
4. Agents	**MARL Engine**	Ray RLlib (PPO)	The "Brains" taking actions	Receives Obs; outputs cluster commands	Multi-Agent PPO Policy
4. Agents	**Referee (Logic)**	Python	Conflict resolution / Tie-breaker	Intercepts Agent B to protect Agent A	Prioritizes Safety over Efficiency
5. Optimizer	**Optuna**	Python / Optuna	Meta-optimization of rewards	Tune alpha, beta, gamma across trials	Bayesian Search for Equilibrium

6. Feedback	**Global Scoreboard**	Python	Final performance tracking	$Combines R_{A,B,C} for Optuna$	$Global_Score = (\alpha \cdot R_A) + (\beta \cdot R_B) + (\gamma \cdot R_C)$

Agent Name	Core Objective	Primary Action	Secondary Actions	Observation Input	Reward Logic (+)	Penalty Logic (-)	Priority
Agent A: Risk	**99.9% Task Survival**	Migrate	Replicate, Throttle	XGBoost 1, Node Health	+10 for risk recovery	-100 for Task Death	**1**
Agent B: Cost	**Min Power & Waste**	Power State	DVFS (Clock), Mem-Balloon, Deep Sleep	XGBoost 2, Idle %	+5 per saved Watts/h	-50 for SLA violation	**3**
Agent C: Load	**Max Throughput**	Admit/ Reject	Queue-TTL, De-prioritize, Resource-Cap	Global Load, Forecast	+5 for job completion	-20 for false rejection	**2**